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DC014 SAINS KOMPUTER SKEMA PRAPSPM SESI 2024/2025

Questions No	Answers	Marks
1(a)	Audio - Audio is the use of speech, music and sound effects. - Produced by vibration, as perceived by the sense of hearing. Graphic - A digital representation of non-text information Animation - Animation is a process of making a static image look like it is moving. - Animation is a series of graphical images displayed one after another to simulate movement. Text - Combination of letters that form a word or sentence that describe a topic which is known as text information. - Text is a basic element of multimedia that combine alphabets, numbers and symbol Video - Video is a series of frames containing pictures and sound played back at a fast rate. Video is a product of recording moving visual images by using a camera or video machine.	J2 1m - element 1m - explain <i>Select any one (1) answer</i>

1(b)	<table><tr><th>Element</th><th>Explanation</th></tr><tr><td>Audio</td><td> • To capture attention: a sound in a multimedia presentation will capture an audience's attention. • To increase the associations the end-users make with the information in their mind (sound of a duck), thus increasing their retention rate. • Audio adds an exciting dimension to a presentation. </td></tr><tr><td>Graphic</td><td> • Convey information more quickly than when using text. • Make complex information easy to understand. • To present information more attractively. • To illustrate ideas through still pictures. • Images - used to add emphasis, direct attention, illustrate concepts and provide background content. • Complementary element - used as background and texture. </td></tr><tr><td>Animation</td><td> • Animation is useful for illustrating concepts which involve movement. • Can help organize thoughts and illustrate processes. • Can graphically simplify complicated concepts and convey complex inter-relationships, which are difficult to visualize. • Inject humor and light-heartedness in a presentation via an animated cartoon. • Can attract audience attention </td></tr><tr><td>Text</td><td> • Effective for conveying basic/fundamental information </td></tr><tr><td>Graphic</td><td> • Videos engage students of all ages and abilities • Videos motivate with authentic content • Videos provide concept/context/real situation to help understanding. </td></tr></table> Correction: Last element, replace graphic with video.	Element	Explanation	Audio	• To capture attention: a sound in a multimedia presentation will capture an audience's attention. • To increase the associations the end-users make with the information in their mind (sound of a duck), thus increasing their retention rate. • Audio adds an exciting dimension to a presentation.	Graphic	• Convey information more quickly than when using text. • Make complex information easy to understand. • To present information more attractively. • To illustrate ideas through still pictures. • Images - used to add emphasis, direct attention, illustrate concepts and provide background content. • Complementary element - used as background and texture.	Animation	• Animation is useful for illustrating concepts which involve movement. • Can help organize thoughts and illustrate processes. • Can graphically simplify complicated concepts and convey complex inter-relationships, which are difficult to visualize. • Inject humor and light-heartedness in a presentation via an animated cartoon. • Can attract audience attention	Text	• Effective for conveying basic/fundamental information	Graphic	• Videos engage students of all ages and abilities • Videos motivate with authentic content • Videos provide concept/context/real situation to help understanding.	J2 1m - keyword 1m - explain <i>Select any one (1) answer</i>
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1(c)	Advantage of Video : I. Video is a very powerful element to convey information, which can incorporate a personal element that other media lack. II. provides a powerful and realistic impact in a multimedia program III. digital video files can be stored like any other files in the computer and the quality of the video can still be maintained IV. Giving users an impression of a speaker's personality. V. Showing things that move. VI. Video clips can be edited easily • To show continuous process • To show complex concept/ action @any suitable answer	J2 1m - keyword 1m - explain <i>Select any two (2) answer</i>												

	Using symbols in a flowchart scheme – each icon represents a particular event.	
2(d)	Hypertext - refers to links in text-based documents. Hypermedia - refers to connecting the hypertext with media such as graphics, sounds and animations. Tabbed Navigation - refers to the clickable area at the top that shows another page or area.	Name (J1) Explain (J1)
3(a)	File processing system	J1
3(b)	Database system - Together, the data and the DBMS, along with the applications that are associated with them.	Name (J1) Describe (J1)
3(c)(i)	- Reducing Data Redundancy - Most data items are stored in only one file, which greatly reduces duplicate data - Improved Data Integrity - the data is correct and consistent in all the databases because the data is visible to multiple users - Sharing of Data - the users can share the data among themselves - Easier access - allows users to access and maintain data, providing they have the necessary privileges. - Reduced development time - easier and faster to develop programs that use the database approach. - Data Security - Only authorised users should be allowed to access the database and their identity should be authenticated using a username and password. - Backup and Recovery - Database Management System automatically takes care of backup and recovery. <i>Any acceptable answer</i>	J1
3(c)(ii)	Data redundancy because data is repeated / occurs/ appears multiple times in each department OR Data Isolation Data is difficult to access by other user from other department because data are kept in separate files in separate department.	J1 Select any one (1) answer

	Any acceptable answer											
4(a)	<table><tr><td>Layers</td><td>Data Hierarchy</td></tr><tr><td>2</td><td>Record</td></tr><tr><td>5</td><td>Byte</td></tr></table>	Layers	Data Hierarchy	2	Record	5	Byte	J1 J1				
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4(b)	<table><tr><td>Data description</td><td>Data type</td></tr><tr><td>Identification card number</td><td>Text</td></tr><tr><td>Yes or No questions</td><td>Boolean</td></tr></table>	Data description	Data type	Identification card number	Text	Yes or No questions	Boolean	J1 J1				
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5(a)	<table><tr><td>Areas</td><td>Explanation</td></tr><tr><td>GPS network</td><td>A satellite navigation system used to determine the ground position of an object.</td></tr><tr><td>Application Monitoring System</td><td>A networking technology to determine the current location or status of an object.</td></tr><tr><td>Multimedia networking</td><td>It is a network application to deliver digital multimedia content to computers. The user usually has to have a computer or smart tv to access this service</td></tr><tr><td>Telephone service</td><td>A voice-grade voice service employing analog signal transmission over copper wires.</td></tr></table>	Areas	Explanation	GPS network	A satellite navigation system used to determine the ground position of an object.	Application Monitoring System	A networking technology to determine the current location or status of an object.	Multimedia networking	It is a network application to deliver digital multimedia content to computers. The user usually has to have a computer or smart tv to access this service	Telephone service	A voice-grade voice service employing analog signal transmission over copper wires.	J2 J2 J1 J1
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5(b)	Fibre Optic Cable Twisted Pair Cable Coaxial Cable	J1 J1 J1										
5(c)	Cellular Radio Network Microwave and Satellite Transmission	J1 J1										
5(d)	<table><tr><td>Label</td><td>Communication device</td></tr><tr><td>A</td><td>Modem</td></tr><tr><td>B</td><td>Router</td></tr></table>	Label	Communication device	A	Modem	B	Router	J1 J1				
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	C	Wireless Access Point	J1
	D	Repeater	J1
	E	Switch/ Hub	J1
5(e)	Network adapter		J1
6(a)	Subject directory / directory search		J1
	A subject directory provides categorised lists of links arranged by subject.		J2
6(b)(i)	Field Search		J1
	Search by adding characteristic to the keywords such as language, region or date		J2
6(b)(ii)	AND		J1
	OR		J1
	NOT		J1
6(c)(i)	Students take classes from a location, often home or work - which is different from the one where the teacher/ delivery of instruction takes place <i>Any acceptable answer.</i>		J2
6(c)(ii)	Online education		J1
	Online testing		J1
	Web based training		J1
			<i>Select any two (2)</i>
6(d)(i)	Internet censorship		J1
6(d)(ii)	The suppression of speech, public communication, or other information over Internet		J2
	<i>Any acceptable answer.</i>		
6(e)	● Regularly Clear Cookies and Cache		J1
	● Avoid Clicking on Suspicious Links		J1
	● Keep Your Software Updated		J1
	● Use Strong, Unique Passwords		J1
	● Enable Two-Factor Authentication (2FA)		J1
	● Use a VPN (Virtual Private Network)		J1
	● Use private browsing		J1
	<i>Any acceptable answer.</i>		<i>Select any two (2)</i>
6(f)	Physical objects with sensors, processing ability, software, and other technologies that connect and exchange data with other devices and systems over the Internet or other communications		J2

	networks <i>Any acceptable answer.</i>	
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